



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

TG: @Chalnaayaaar NEET (UG)

MINOR

16-07-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEB-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E6

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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Your Target is to secure Good Rank in Pre-Medical 2024

Topic : CAPACITOR, MAGNETIC EFFECT OF CURRENT (UPTO APPLICATION OF BSL), INTERFERENCE OF SOUND TO WAVE MOTION (COMPLETE), NLM, [BACK UNIT] : CAPACITOR AND GRAVITATION

SECTION-A (PHYSICS)

1. In a resonance tube, the first resonance with a tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is (in Hz) :-

(1) 500 (2) 300 (3) 330 (4) 165

2. A uniform string of length L and mass M is fixed at both ends under tension T , Then it can vibrate with frequency given by the formula.

(1) $f = \frac{1}{2} \sqrt{\frac{T}{ML}}$

(2) $f = \frac{1}{2L} \sqrt{\frac{T}{M}}$

(3) $f = \frac{1}{2} \sqrt{\frac{T}{M}}$

(4) $f = \frac{1}{2} \sqrt{\frac{M}{LT}}$

3. Two waves having equation :

$$x_1 = a \sin(\omega t + \phi_1) \quad x_2 = a \sin(\omega t + \phi_2)$$

If in the resultant wave the frequency and amplitude remains equals to amplitude of superimposing waves. Then phase diff. between them.

(1) $\pi/6$ (2) $2\pi/3$

(3) $3\pi/4$ (4) $\pi/4$

4. Two waves of wave length 2 m and 2.02 m respectively moving with the same velocity and superimpose to produce 2 beats per sec. The velocity of the waves is :

(1) 400.0 m/s

(2) 402 m/s

(3) 404 m/s

(4) 406 m/s

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5. The fundamental frequency of a closed pipe is 220 Hz. If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is :

(1) 220 Hz

(2) 440 Hz

(3) 880 Hz

(4) 1760 Hz

6. A bus is moving with a velocity of 5m/s towards a huge wall. The driver sounds a horn of frequency of 165 Hz. If the speed of sound in air is 335 m/s, number of beats heard by a passenger on bus will be :-

(1) 6 (2) 5 (3) 3 (4) 4

7. Two concentric coils each of radius equal to 2π cm are placed at right angles to each other 3 amp. & 4 amp are the current flowing in each coil. Magnetic field at the centre of the coil will be :-

(1) 7×10^{-5} T

(2) 5×10^{-5} T

(3) 10^{-5} T

(4) 12×10^{-5} T

8. An equilateral triangular loop of wire of side ℓ carries a current i . The magnetic field produced at the circum center of the loop is :-

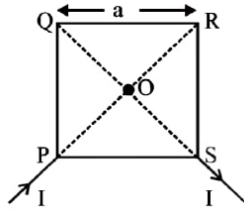
(1) $\frac{\mu_0}{4\pi} \frac{3\sqrt{3}i}{\ell}$

(2) $\frac{\mu_0}{4\pi} \frac{9i}{\ell}$

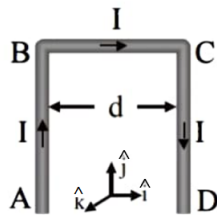
(3) $\frac{\mu_0}{4\pi} \frac{18i}{\ell}$

(4) $\frac{\mu_0}{4\pi} \frac{6i}{\ell}$

9. In a square loop PQRS made with a wire of uniform cross-section as shown in Fig. current I enters from point P and leaves from point S. The magnitude of magnetic field induction at the centre O of the square is :-



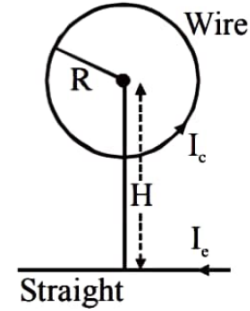
- (1) $\frac{\mu_0}{4\pi} \frac{2\sqrt{2}I}{a}$ (2) $\frac{\mu_0}{4\pi} \frac{4\sqrt{2}I}{a}$
 (3) $\frac{\mu_0}{\pi} \frac{4\sqrt{2}I}{a}$ (4) Zero
10. A length L of wire carries a steady current I . It is bent first to form a plane circular coil of one turn. The same length is now bent more sharply to give a double loop of smaller radius. The magnetic field at the centre caused by the same current is :-
- (1) A quarter of its first value
 (2) Unaltered
 (3) Four times of its first value
 (4) A half of its first value
11. AB and CD are long straight conductors, distance d apart, carrying a current I . The magnetic field at the midpoint of BC is :-



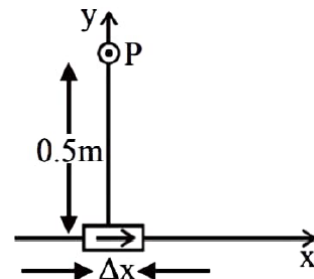
- (1) $\frac{-\mu_0 I}{2\pi d} \hat{k}$
 (2) $\frac{-\mu_0 I}{\pi d} \hat{k}$
 (3) $\frac{-\mu_0 I}{4\pi d} \hat{k}$
 (4) $\frac{-\mu_0 I}{8\pi d} \hat{k}$

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12. Circular loop of a wire and a long straight wire carry currents I_c and I_e , respectively as shown in figure. Assuming that these are placed in the same plane. The magnetic fields will be zero at the centre of the loop when the separation H is :-



- (1) $\frac{I_e R}{I_c \pi}$ (2) $\frac{I_c R}{I_e \pi}$
 (3) $\frac{\pi I_c}{I_e R}$ (4) $\frac{I_e \pi}{I_c R}$
13. An element $d\ell = dx \hat{i}$ is placed at the origin and carries a large current $I = 10$ A. What is the magnetic field on the y -axis at a distance of 0.5 m. $\Delta x = 1$ cm :-

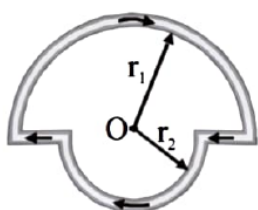


- (1) 4×10^{-8} T in $+z$ direction
 (2) 4×10^{-8} T in $-z$ direction
 (3) 2×10^{-8} T in $+z$ direction
 (4) 2×10^{-8} T in $-z$ direction
14. Dimension of μ_0 is :-
- (1) $M^1 L^1 T^{-2} A^{-2}$
 (2) $M^1 L^2 T^{-2} A^{-2}$
 (3) $M^1 L^1 T^{-2} A^{-1}$
 (4) None

15. A long wire carries a current of 90A from east to west direction. Magnetic field due to current at a point which is 3 m just below the wire :-

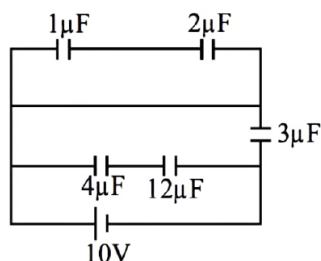
- (1) 12×10^{-6} T, towards north
- (2) 6×10^{-6} T, towards south
- (3) 12×10^{-6} T, towards south
- (4) 6×10^{-6} T, towards north

16. In the figure shown there are two semicircles of radii r_1 and r_2 in which a current i is flowing. The magnetic induction at the centre O will be :-



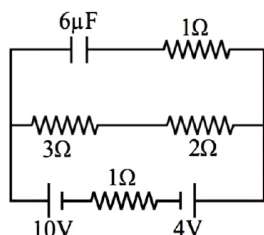
- (1) $\frac{\mu_0 i}{r}(r_1 + r_2)$
- (2) $\frac{\mu_0 i}{4}(r_1 - r_2)$
- (3) $\frac{\mu_0 i}{4} \left(\frac{r_1 + r_2}{r_1 r_2} \right)$
- (4) $\frac{\mu_0 i}{4} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$

17. Find out the value of net charge that flows through the given circuit :



- (1) 40 μ C
- (2) 20 μ C
- (3) 120 μ C
- (4) 60 μ C

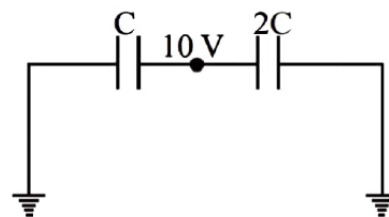
18. Find out charge on capacitor in steady state given in figure :



- (1) 30 μ C
- (2) 20 μ C
- (3) 40 μ C
- (4) 60 μ C

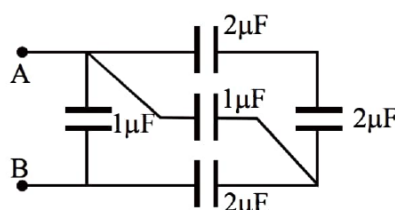
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19. In the circuit in figure $C = 6 \mu$ F. The charge stored in the capacitor of capacity C is :-



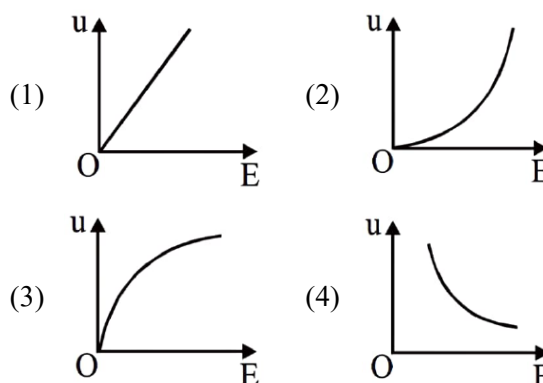
- (1) Zero
- (2) 90 μ C
- (3) 40 μ C
- (4) 60 μ C

20. The effective capacity between A and B in Fig. will be :-



- (1) 1 μ F
- (2) 2 μ F
- (3) $\frac{2}{3} \mu$ F
- (4) 4 μ F

21. If u is the energy density and E is the electric field then correct graph is :-

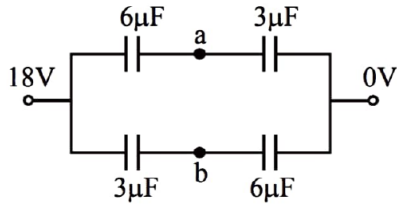


22. Small drops of the same size are charge to V volt each.

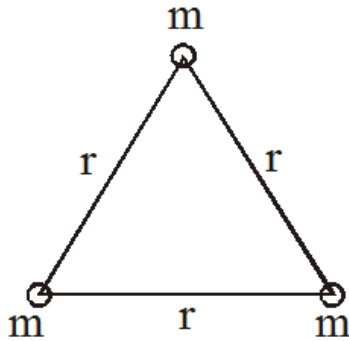
If 27 such drops coalesce to form a single large drop then its potential will be :-

- (1) 27 V
- (2) 9 V
- (3) 3 V
- (4) 243 V

23. Find out ($V_a - V_b$) in given circuit :-



- (1) 6V (2) 8V
(3) 4V (4) 12V
24. Three equal mass (m) are placed at vertex of an equilateral triangle of side r . work required to double the separation between masses will be :-



- (1) $\frac{Gm^2}{r}$ (2) $\frac{3Gm^2}{r}$
(3) $\frac{3}{2} \frac{Gm^2}{r}$ (4) None
25. A body is projected up from the surface of the earth with a velocity equal to $\frac{3}{4}$ th of escape velocity for earth's surface. If R be the radius of the earth, the maximum height it reaches is :-
- (1) $\frac{3R}{10}$
(2) $\frac{9R}{7}$
(3) $\frac{8R}{5}$
(4) $\frac{9R}{5}$
26. If the density of a small planet is the same as that of the earth, while the radius of the planet is 0.2 times that of the earth, the gravitational acceleration on the surface of that planet is :-
- (1) 0.2g (2) 0.4g (3) 2g (4) 4g

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27. A geostationary satellite is orbiting the Earth at a height of $6R$ above the surface of the earth, R being the radius of the Earth. The time period of another satellite, at a height of $2.5R$ from the surface of the Earth, is

- (1) $6\sqrt{2}$ hour
(2) 6 hour
(3) $5\sqrt{2}$ hour
(4) 10 hour

28. Find the depth where g remains 60% of the value at the surface. [$R = 6400$ Km]

- (1) 3840 Km
(2) 1920 Km
(3) 2560 Km
(4) 1280 Km

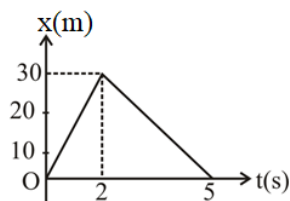
29. A satellite is moving in an elliptical path around a planet

- (a) Angular momentum
(b) Kinetic energy
(c) Potential energy
(d) Mechanical energy

Which of the above is/are conserved for a satellite.

- (1) abc (2) ac (3) ad (4) bcd

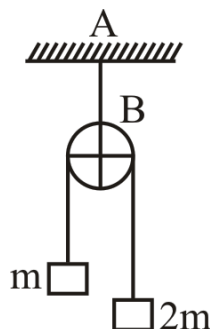
- 30.



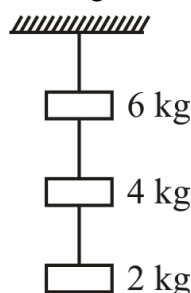
A particle of mass 20 g is moving along x-axis according to given position-time graph then find impulse at $t = 2$ s.

- (1) -0.1 kg m/s
(2) -0.5 kg m/s
(3) 0.1 kg m/s
(4) 0.5 kg m/s

31. In the arrangement shown in the figure, the pulley is massless. Neglecting friction on the contact surface, the force exerted by the supporting rope AB on the ceiling is :-



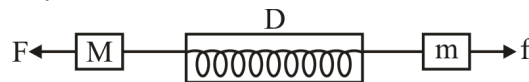
- (1) $6mg$
 (2) $3mg$
 (3) $4mg$
 (4) None of these
32. A monkey is descending from the branch of a tree with a constant acceleration. If the breaking strength of the branch is 75% of the weight of the monkey, the minimum acceleration with which the monkey can slide down without breaking the branch is :-
- (1) g (2) $3g/4$
 (3) $g/2$ (4) $g/4$
33. Three masses of 6 kg, 4 kg and 2 kg are attached to a rigid support as shown in figure. If the string attached to the support breaks and the system falls freely, then the tension in the string connecting 4 kg and 2 kg mass is :-



- (1) zero (2) 8 kg wt
 (3) 12 kg wt (4) 6 kg wt

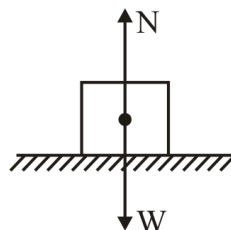
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34. A dynamometer D is attached to two bodies of masses $M = 6 \text{ kg}$ and $m = 4 \text{ kg}$. Forces $F = 20 \text{ N}$ and $f = 10 \text{ N}$ are applied to the masses as shown. The dynamometer reads :-



- (1) 10 N (2) 20 N (3) 6 N (4) 14 N

- 35.



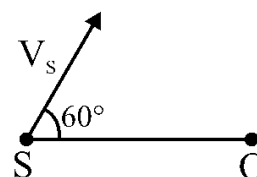
Statement-I :- The block is at rest under the two forces so resultant of both the force is zero.

Statement-II :- The resultant of forces on the block is zero, so it is at rest.

- (1) Both Statement-I and Statement-II are incorrect.
 (2) Statement-I is correct but Statement-II is incorrect.
 (3) Statement-I is incorrect but Statement-II is correct
 (4) Both Statement-I and Statement-II are correct.

SECTION-B (PHYSICS)

36. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 m/s at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air = 330 m/s) is :-



- (1) 97 Hz (2) 100 Hz
 (3) 103 Hz (4) 106 Hz

37. Beats are produced by two waves :

$$y_1 = a \sin 1000\pi t \text{ and } y_2 = a \sin 998\pi t$$

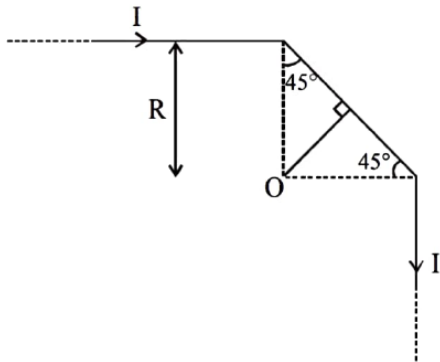
The number of beats heard/sec is :-

- (1) 0 (2) 2
(3) 1 (4) 4

38. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are: (velocity of sound = 340 ms^{-1})

- (1) 4 (2) 5
(3) 7 (4) 6

39.



A long infinite current carrying wires bent in the shape as shown in figure. The magnetic induction at point O is :-

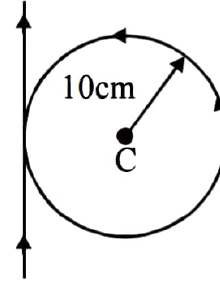
- (1) $\frac{\mu_0 I}{\pi R} \otimes$
(2) $\frac{\mu_0 I}{2\pi R} \otimes$
(3) $\frac{\mu_0 I}{\pi R} \odot$
(4) $\frac{\mu_0 I}{2\pi R} \odot$

40. A straight section PQ of a circuit lies along the X-axis from $x = -\frac{a}{2}$ to $x = \frac{a}{2}$ and carries a steady current i . The magnetic field due to the section PQ at a point $X = +a$ will be :-

- (1) Proportional to a
(2) Proportional to a^2
(3) Proportional to $1/a$
(4) Zero

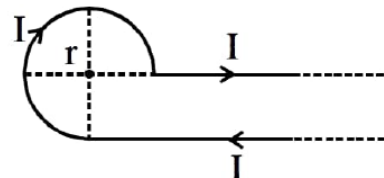
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41. A long, straight wire is turned into a loop of radius 10 cm (see figure). If a current of 8 amperes is passed through the loop, then the value of the magnetic field and its direction at the centre C of the loop shall be close to :-



- (1) 5.0×10^{-5} Newton/(amp-meter), upward
(2) 3.4×10^{-5} Newton/(amp-meter), upward
(3) 1.6×10^{-5} Newton/(amp-meter), downward
(4) 1.6×10^{-5} Newton/(amp-meter), upward

42. Current I is flowing in conductor shaped as shown in the figure. The radius of the curved part is r and the length of straight portion is very large. The value of the magnetic field at the centre O will be :-



- (1) $\frac{\mu_0 I}{4\pi r} \left(\frac{3\pi}{2} + 1 \right)$
(2) $\frac{\mu_0 I}{4\pi r} \left(\frac{3\pi}{2} - 1 \right)$
(3) $\frac{\mu_0 I}{4\pi r} \left(\frac{\pi}{2} + 1 \right)$
(4) $\frac{\mu_0 I}{4\pi r} \left(\frac{\pi}{2} - 1 \right)$

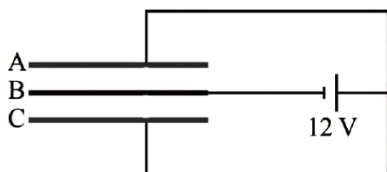
43. Two capacitors marked ($12\mu\text{F}$, 200V), ($24\mu\text{F}$, 200V) are connected in series. The combination acts as a capacitor with breakdown voltage :-

- (1) ($36\mu\text{F}$, 200V) (2) ($8\mu\text{F}$, 400V)
(3) ($8\mu\text{F}$, 300V) (4) ($36\mu\text{F}$, 400V)

44. A number of condensers, each of the capacitance $1 \mu\text{F}$ and each one of which gets punctured if a potential difference just exceeding 500 V is applied, are provided. An arrangement suitable for giving capacitance of $2 \mu\text{F}$ across which 3000 volt may be applied requires at least :-

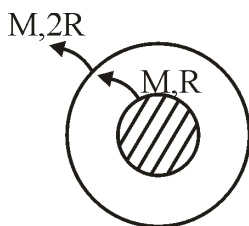
- (1) 6 component capacitors
- (2) 12 component capacitors
- (3) 72 component capacitors
- (4) 2 component capacitors

45. Three plates A,B,C each of area 50 cm^2 have separation 3 mm between A and B and 3 mm between B and C. The energy stored when the plates are fully charged is :-



- (1) $1.6 \times 10^{-9} \text{ J}$
- (2) $2.1 \times 10^{-9} \text{ J}$
- (3) $5 \times 10^{-9} \text{ J}$
- (4) $7 \times 10^{-9} \text{ J}$

46. A solid sphere (M, R) is placed inside a spherical shell ($M, 2R$). Find the potential at the centre of the system.



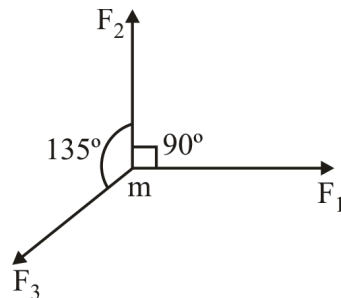
- (1) $-\frac{2GM}{R}$
- (2) $-\frac{3GM}{R}$
- (3) $-\frac{GM}{2R}$
- (4) $-\frac{GM}{R}$

47. The gravitational potential in a region is given as $V = x^2y$. Find the magnitude of gravitational field intensity $\left(\frac{\text{N}}{\text{Kg}}\right)$ at $(2, 1)$.

- (1) $2\sqrt{2}$
- (2) 2
- (3) 4
- (4) $4\sqrt{2}$

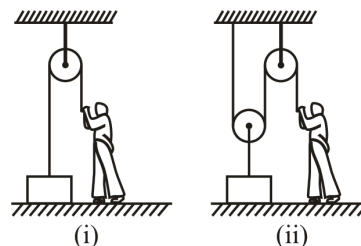
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48. When a force F acts on a body of mass m , the acceleration produced in the body is a . If three equal forces $F_1 = F_2 = F_3 = F$ act on the same body as shown in figure. The acceleration produced is :-



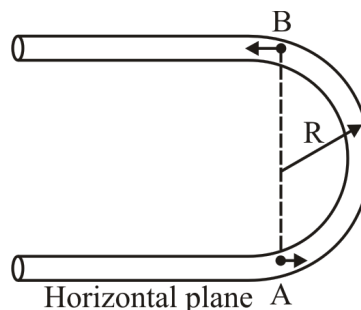
- (1) $(\sqrt{2} - 1)a$
- (2) $(\sqrt{2} + 1)a$
- (3) $\sqrt{2}a$
- (4) a

49. In the figure shown, a person wants to raise a block lying on the ground to a height h . In both the cases if time required is the same then in which case he has to exert more force. Assume pulleys and strings are light.



- (1) (i)
- (2) (ii)
- (3) Same in both
- (4) Cannot be determined

50. A particle of mass ' m ' is moving with constant speed ' V ' throughout the motion then average force on this particle from A to B.



- (1) $\frac{2mV}{\pi R}$
- (2) $\frac{2mV^2}{\pi R}$
- (3) $\frac{mV^2}{\pi R}$
- (4) $\frac{mV^2}{R}$

Topic : d & f-BLOCK, METALLURGY, ISOMERISM (OPTICAL AND CONFORMATIONAL), GOC UPTO COMPLETION, MOLE CONCEPT, CHEMICAL EQUILIBRIUM, [BACK UNIT] : CHEMICAL BONDING

SECTION-A (CHEMISTRY)

51. Which of the following hydrated ions are expected to be coloured ?

1. Cu^+ 2. Ti^{4+} 3. Co^{2+} 4. Fe^{2+} 5. Zn^{+2}

(1) 1, 2, 3, 4 (2) 3, 4

(3) 2, 3, 5 (4) 1, 2

52. Which of the following species do not exist ?

(1) VI_5 (2) CoF_3

(3) VF_5 (4) CrF_6

53. Which of the following statement(s) is correct when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ?

(I) A deep red vapour is evolved.

(II) The vapour when passed through NaOH solution, gives a yellow solution.

(III) Chromyl chloride is formed.

(1) (I) and (II)

(2) (I) and (III)

(3) (III) and (I)

(4) (I), (II) and (III)

54. Match the Following :-

	List-I		List-II
(P)	$[\text{M}(\text{CN})_2]^-$	(A)	Ni (during refining Metal)
(Q)	$[\text{M}(\text{CO})_4]$	(B)	Ti (during refining Metal)
(R)	Ml_4	(C)	Pb (during roasting of ore)
(S)	MO	(D)	Ag (during leaching of ore)

(1) P-D, Q-A, R-B, S-C

(2) P-D, Q-B, R-A, S-C

(3) P-C, Q-A, R-B, S-D

(4) P-A, Q-B, R-D, S-C

TG: @Chalnaayaaar

55. **Assertion (A):** In the extraction of Iron metal Hematite, Silicon dioxide is added as a flux.

Reason (R): Lime stone decomposed to calcium oxide which removes the impurity of ore.

(1) Both Assertion (A) and Reason (R) are correct statements Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are correct statements but Reason (R) is the not the correct explanation of Assertion (A).

(3) Assertion (A) is correct, but Reason (R) is wrong

(4) Assertion (A) is wrong, but Reason (R) is correct statement

56. Match column-I with column-II :

(P)	ClO_3^+	(1)	sp^3 with no l.p
(Q)	ClO_3^-	(2)	sp^3d
(R)	ClF_3	(3)	sp^2
(S)	SO_2Cl_2	(4)	sp^3d^2
		(5)	sp^3 with one l.p.

(1) P-2, Q-3, R-4, S-5

(2) P-3, Q-5, R-2, S-1

(3) P-3, Q-2, R-4, S-1

(4) P-1, Q-2, R-3, S-4

57. Identify correct order of stability of following molecules:

(1) $\text{O}_2 > \text{O}_2^+ > \text{O}_2^- > \text{O}_2^{-2}$

(2) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{-2}$

(3) $\text{O}_2^- > \text{O}_2 > \text{O}_2 > \text{O}_2^{-2}$

(4) $\text{O}_2^- > \text{O}_2 > \text{O}_2^{-2} > \text{O}_2^+$

58. Which of the following species can't exist :-

(1) NF_3 (2) $[\text{SiF}_6]^{-2}$

(3) IH_5 (4) BrF_3

59. Which of the following relation is correct for bond angles in given structure ?



- (1) $x = y < z$ (2) $x = y = z$
 (3) $x = y > z$ (4) $x > y > z$
60. Maximum number of 90° bond angles are present in :-
 (1) SF_4 (2) IF_5 (3) PH_3 (4) IF_7
61. Which of the following molecule is not cyclic ?
 (1) CrO_5 (2) C_3O_2 (3) S_3O_9 (4) $\text{H}_3\text{P}_3\text{O}_9$
62. Correct order boiling point is :-
 (1) $\text{H}_2 < \text{He}$ (2) $\text{ICl} > \text{Br}_2$
 (3) $\text{CH}_3\text{Cl} < \text{CCl}_4$ (4) $\text{He} > \text{Xe}$
63. Select correct statement regarding given reaction :-
 $\text{LiNO}_3 \xrightarrow{\Delta} \text{X(solid)} + \text{Y(gas)} + \text{O}_2$
 (1) Y is paramagnetic and colourless.
 (2) Y is diamagnetic and colourless
 (3) Y is diamagnetic and coloured
 (4) Y is paramagnetic and coloured
64. Match list-I with list-II :-

List-I (Metals)		List-II (Metal ores)	
(A)	Tin	(I)	Silver glance
(B)	Magnesium	(II)	Siderite
(C)	Silver	(III)	Dolomite
(D)	Iron	(IV)	Cassiterite

Choose the correct option given below :-

- (1) A-IV, B-III, C-I, D-II
 (2) A-II, B-IV, C-I, D-III
 (3) A-III, B-IV, C-I, D-II
 (4) A-II, B-III, C-I, D-IV

TG: @Chalnaayaaar

65. **Assertion (A)** : Many trivalent lanthanoid ions are coloured both in the solid state and in aqueous solutions.

Reason (R) : Colour of ions may be attributed to the presence of unpaired f-electrons.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (2) **Assertion** is true but **Reason** is false.
 (3) **Assertion** is false but **Reason** is true.
 (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
66. The volume occupied by one molecule of water : (density = 1 g/ml) is -
 (1) 18 mL (2) $\frac{18}{N_A}$ mL
 (3) $\frac{85.55}{N_A}$ mL (4) $\frac{N_A}{22400}$ mL
67. 20 g of an ideal gas occupies 5.6 L at 1 atm and 273 K. What is the molecular weight of gas :
 (1) 64 (2) 80
 (3) 96 (4) None
68. The simplest formula of compound containing 50% of element X (Atomic mass = 10) and 50% of element Y (Atomic mass = 20) is :-
 (1) XY (2) X_2Y
 (3) XY_3 (4) X_2Y_3
69. Calculate the number of electrons in 18 mL of water ($d = 1 \text{ g/mL}$) ?
 (1) 10 (2) 18
 (3) $10 N_A$ (4) $18 N_A$
70. Equal mass of oxygen, hydrogen and methane are taken in a container in identical conditions. What is the ratio of their moles :-
 (1) 1 : 16 : 1 (2) 1 : 16 : 2
 (3) 8 : 1 : 8 (4) 16 : 1 : 8

71. Find volume of $\text{CO}_2(\text{g})$ produced at STP on decomposition of 80% pure 100g $\text{CaCO}_3(\text{s})$:

- (1) 22.4 L (2) 11.2 L
(3) 17.9 L (4) 14.2 L

72. If expression for equilibrium constant for a gaseous reaction is :-

$$K_C = \frac{[\text{SO}_3]_{\text{eq}}^2}{[\text{SO}_2]_{\text{eq}}^2 [\text{O}_2]_{\text{eq}}}$$

Then the balanced chemical equation would be :-

- (1) $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$
(2) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$
(3) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
(4) None

73. For a reaction $\frac{1}{2}\text{H}_2 + \frac{1}{2}\text{I}_2 \rightleftharpoons \text{HI}$, equilibrium constant is K. Then what will be the equilibrium constant for the reaction $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$:-

- (1) $\frac{1}{K^2}$
(2) K^2
(3) $\frac{1}{\sqrt{K}}$
(4) K

74. The equilibrium constant (K_p) for the reaction $\text{PCl}_{5(\text{g})} \rightleftharpoons \text{PCl}_{3(\text{g})} + \text{Cl}_{2(\text{g})}$ is 16. If the volume of the container is reduced to half of its original volume, the value of K_p for the reaction at the same temperature will be :-

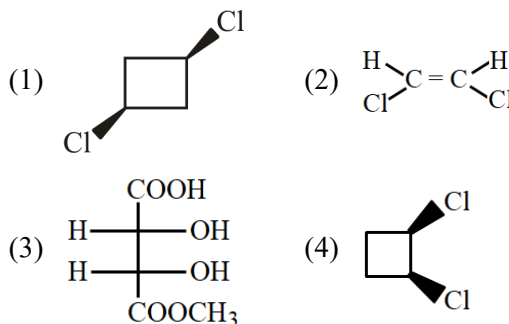
- (1) 32 (2) 64
(3) 16 (4) 4

75. For which of the following equilibrium $K_c > K_p$

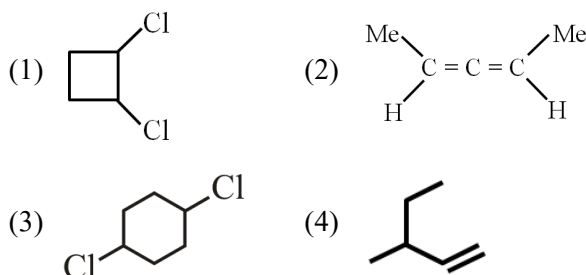
- (1) $2\text{NO}_{(\text{g})} \rightleftharpoons \text{N}_{2(\text{g})} + \text{O}_{2(\text{g})}$
(2) $\text{SO}_{2(\text{g})} + \text{NO}_{2(\text{g})} \rightleftharpoons \text{SO}_{3(\text{g})} + \text{NO}_{(\text{g})}$
(3) $\text{A}_{(\text{g})} + 3\text{B}_{(\text{g})} \rightleftharpoons 4\text{C}_{(\text{g})} + 5\text{D}_{(\text{g})}$
(4) $\text{PCl}_{3(\text{g})} + \text{Cl}_{2(\text{g})} \rightleftharpoons \text{PCl}_{5(\text{g})}$

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76. Which of the following is meso compound ?



77. Which of the following will show optical as well as Geometrical isomerism ?



78. Which of the following - I effect strength is correct ?

- (1) $-\text{F} > -\text{C} \equiv \text{CH} > -\text{OH}$
(2) $-\text{F} > -\text{OH} > -\text{NH}_2 > -\text{C} \equiv \text{CH}$
(3) $-\text{F} > -\text{OH} > -\text{C} \equiv \text{CH} > -\text{NH}_2$
(4) $-\text{F} > -\text{NH}_2 > -\text{OH} > -\text{Ph}$

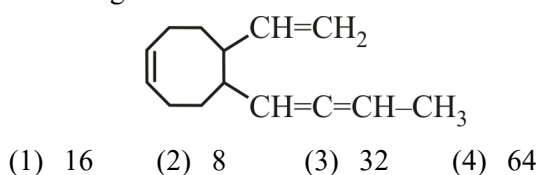
79. Which of the following is not a Electrophile ?

- (1) NO_2^+ (2) SO_3
(3) NH_4^+ (4) Cl^+

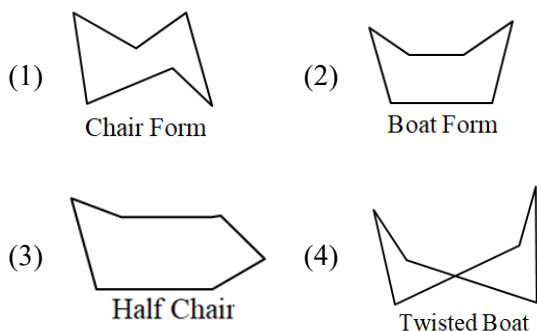
80. Which of the following can show conformational isomerism ?

- (1) $\text{CH}_3 - \text{Cl}$ (2) H_2O
(3) NH_3 (4) $\text{CH}_3 - \text{CH}_3$

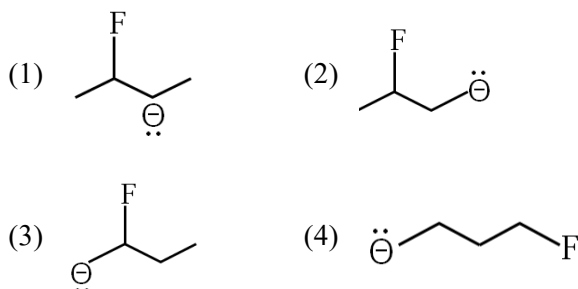
81. Calculate the number of stereoisomers from following ?



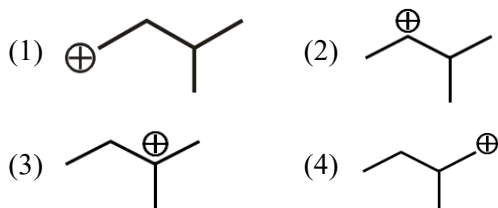
82. Which of the following conformation of cyclohexane has both POS and COS ?



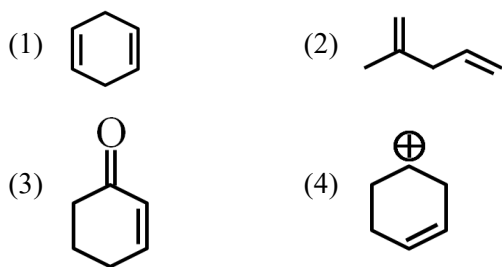
83. Which is most stable



84. Most stable carbocation is



85. Which molecule is conjugated ?



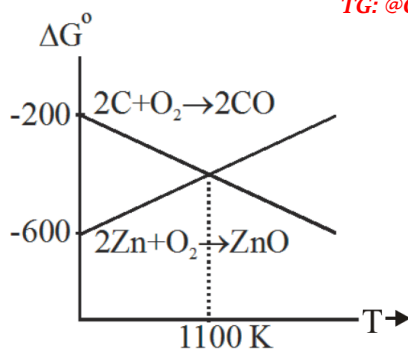
SECTION-B (CHEMISTRY)

86. False about KMnO_4 is

- (1) it is coloured due to charge transfer
- (2) it shows temperature dependent weak paramagnetism
- (3) Hybridisation of Mn is dsp^2
- (4) All Mn - O bonds are identical

TG: @Chalnaayaaar

87.



To make $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$

reduction process spontaneous, the temperature should be

- (1) 1100 K
- (2) >1100 K
- (3) <500 K
- (4) <1100 K

88. Which of the following statement is incorrect :-

- (1) $\text{AgCl} < \text{AgF}$ (Solubility)
- (2) $\text{KCl} < \text{KI}$ (Solubility in a acetone)
- (3) $\text{BeC}_2\text{O}_4 < \text{BaC}_2\text{O}_4$ (Solubility)
- (4) $\text{CaCrO}_4 > \text{BaCrO}_4$ (Solubility)

89. Incorrect statement for C_6H_6 and $\text{B}_3\text{N}_3\text{H}_6$ is :

- (1) They are isoelectronic
- (2) They are isostructural
- (3) Reactivity of C_6H_6 is more than $\text{B}_3\text{N}_3\text{H}_6$ in nonpolar solvent
- (4) Both have $(4n + 2)\pi$ electron, hence antiaromatic

90. Which of the following reaction can not be used during reduction ?

- (1) $\text{ZnO} + \text{Cu} \xrightarrow{\Delta} \text{Zn} + \text{CuO}$
- (2) $\text{TiCl}_4 + \text{Mg} \xrightarrow{\Delta} \text{Ti} + \text{MgCl}_2$
- (3) $[\text{Ag}(\text{CN})_2]^- + \text{Zn} \xrightarrow{\Delta} [\text{Zn}(\text{CN})_4]^{2-} + \text{Ag}$
- (4) $\text{Cu}_2\text{O} + \text{H}_2 \xrightarrow{\Delta} \text{Cu} + \text{H}_2\text{O}$

Topic : GENETICS (POI) : CHROMOSOMAL THEORY OF INHERITANCE TO HUMAN GENETICS, BIOTECHNOLOGY (PART-I), DIGESTION AND ABSORPTION, EXCRETORY PRODUCTS AND THEIR ELIMINATION, [BACK UNIT] : HUMAN HEALTH AND DISEASE

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

- 101.** Which of the following component is a part of "2nd line of defence" in Immunity ?
- (1) Mucosa (2) Immunoglobulins
(3) T-killer cell (4) NK-cell
- 102.** Which of the following immune response is responsible for graft rejection ?
- (1) Physical barriers of innate immunity
(2) Cellular barriers of innate immunity
(3) Humoral immune response
(4) Cell mediated immune response
- 103.** When a patient is suffering from the symptoms of hypersensitivity due to over secretion of histamine and serotonin like substances what you expect in his body?
- (1) Deficiency of B-cells
(2) Deficiency of T-cells
(3) Excess activation of mast cells and basophils
(4) Deficiency of acidophils
- 104.** A disorder in which deficiency of enzyme, tyrosinase causes skin & iris of eye to become melanin deficient is :-
- (1) Alkaptonuria
(2) Thalassemia
(3) Albinism
(4) Pernicious anemia
- 105.** XXY genotype with 47 chromosomes is related to :-
- (1) Turner's syndrome
(2) Klinefelter's syndrome
(3) Super male
(4) Super female

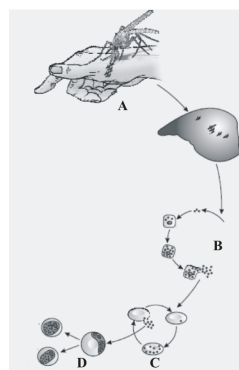
TG: @Chalnaayaaar

- 106.** Which of the following is a 'vector borne bacterial disease'?
- (1) Filariasis (2) Amoebiasis
(3) Chikungunya (4) Plague

- 107.** Select the correct set of only viral diseases.

- (1) Elephantiasis, Dysentery, Chikungunya
(2) Ascariasis, Plague, Hepatitis-B
(3) Chikungunya, Hepatitis-B, Genital warts
(4) Amoebiasis, Hepatitis-B, Chikungunya

- 108.**



Above diagram shows life cycle of plasmodium. Which of the following is correct description of A,B,C and D :-

- (1) A- gametocyte stage infects human body
(2) B- plasmodium multiplying by sexual method of reproduction
(3) C- human RBC are ruptured and releases a toxin hemozoin
(4) D- further development ahead is possible only in female culex mosquito
- 109.** Internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage are the common symptoms of
- (1) Amoebiasis (2) Elephantiasis
(3) Typhoid (4) Ascariasis

110. Which of the following is **not** a sexually transmitted disease ?

- (1) Acquired Immuno Deficiency Syndrome (AIDS)
- (2) Trichomoniasis
- (3) Encephalitis
- (4) Syphilis

111. Which of the following statement is false for AIDS:-

- (1) It can only be tackled by the society and medical fraternity acting together, to prevent the spread of disease
- (2) It has no cure and prevention is best option
- (3) More often it spreads through conscious behaviour pattern
- (4) Treatment of AIDS with anti retroviral drugs is completely effective

112. After getting into the body of the person HIV virus enters into macrophages where RNA genome of the virus replicates to form viral DNA with the help of ?

- (1) DNA polymerase
- (2) Transcriptase
- (3) Reverse transcriptase
- (4) Protease

113. Which one of the following techniques is safest for the detection of cancers ?

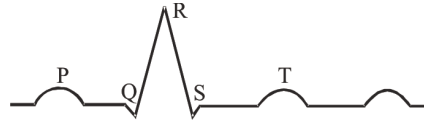
- (1) Radiography (X-ray)
- (2) Computed tomography (CT)
- (3) Histopathological studies
- (4) Magnetic resonance imaging (MRI)

114. **Assertion** : Cocain has a potent stimulating action on central nervous system.

Reason : It interferes with the transport of the Dopamine.

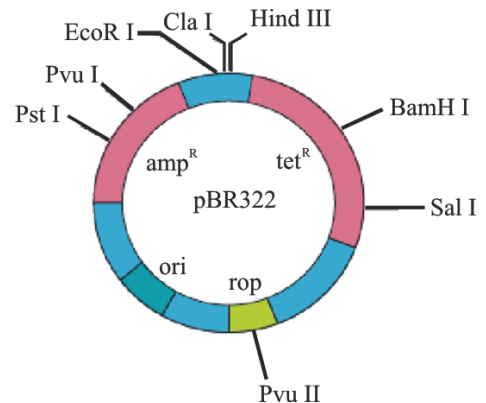
- (1) Both (A) and (R) are correct, but (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct, but (R) is the not correct explanation of (A).
- (3) Both (A) and (R) incorrect.
- (4) (A) is correct and (R) is incorrect.

115. Find out the correct identification for PQRST in the given graph :-



- (1) Peak of R-represent Tricuspid and bicuspid opening
- (2) T-wave ventricular depolarization
- (3) P-wave Depolarisation of atria
- (4) QRS-wave represents ventricular repolarisation

116. Identify the correct option about figure :-



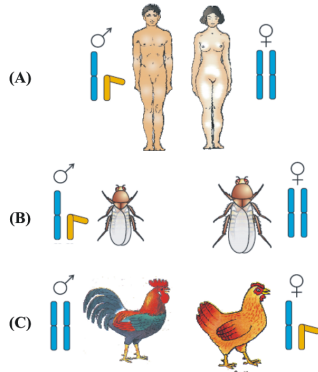
- (1) Eco RI is a restriction site of an enzyme which produces blunt ends
- (2) Hind III is a restriction site of an enzyme which is obtained from E.coli.
- (3) Ori is origin of replication.
- (4) tet^R is gene which is responsible for antibiotic sensitivity.

117. All good cloning vector should have all except:-
- (1) Origin of replication
 - (2) Many recognition sites for single restriction enzyme
 - (3) Selectable markers
 - (4) Ori support high copy no.
118. In r-DNA technology or genetic engineering elution means –
- (1) Remove the DNA from centrifuge tube after centrifugation
 - (2) The separated band of DNA are cut out from the gel and extracted from the gel piece
 - (3) Separation of the recombinant protein from recombinant cell
 - (4) Insertion of recombinant DNA into the host cell
119. Arrange of steps of rDNA technology in a correct order :-
- (I) Insertion of rDNA into host cell
 - (II) Restriction Enzyme digestion
 - (III) Obtaining foreign gene product
 - (IV) Isolation of DNA
 - (V) Amplification of gene of Interest
- (1) IV, II, V, I, III (2) I, II, III, IV, V
 - (3) V, IV, III, II, I (4) II, III, I, V, IV
120. **Assertion** : Unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant vector molecule cannot be created.
Reason : When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky ends'.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - (3) Assertion is True but the Reason is False.
 - (4) Both Assertion & Reason are False.

TG: @Chalnaayaaar

121. Plasmid of which of the following bacterium is used for transfer for foreign gene in plant cells?
- (1) *Agrobacterium tumefaciens*
 - (2) *Escherichia coli*
 - (3) *Bacillus thuringensis*
 - (4) *Salmonella typhimurium*
122. To isolate DNA from the plant cells we have to break the wall this is done by :-
- (1) Lysozyme (2) Cellulase
 - (3) Chitinase (4) Invertase
123. During isolation of DNA, addition of which of the following causes precipitation of purified DNA?
- (1) Chilled ethanol
 - (2) Ribonuclease enzyme
 - (3) DNA polymerase
 - (4) Proteases
124. Who was noted that the behaviour of chromosomes was parallel to the behaviour of genes ?
- (1) Sutton and Boveri (2) Griffith
 - (3) Harshey and Chase (4) Mendel
125. Position of genes on chromosomes was mapped for the first time by :
- (1) Muller (2) Alfred Strurtevant
 - (3) C.Bridges (4) Bateson
126. A colour blind man marries a daughter of normal woman and colour blind man. The ratio of carrier daughter, colour blind daughters, normal sons & colour blind sons born to this couple will be :-
- (1) 1 : 0 : 1 : 0 (2) 2 : 1 : 1 : 2
 - (3) 1 : 1 : 1 : 1 (4) 1 : 2 : 2 : 1

127. The figure below shows three types of sex determination. Select the option giving correct identification.



- (1) C- ZZ males, ZW females
- (2) A - XO Males, XX females
- (3) B - XY females, XX males
- (4) C - ZZ females, XX males

128. Which of the following symbols and its representation is incorrect :-

- (1) = Normal male
- (2) = Carrier female of sex linked dominant disease
- (3) = Death of female individual
- (4) = Sex unspecified

129. Malignant malaria is caused by :-

- (1) Plasmodium malaria
- (2) Plasmodium falciparum
- (3) Plasmodium Vivax
- (4) Plasmodium Ovale

130. Which of the following disease can be caused by inhaling droplets/aerosol released by infected person ?

- (1) Dengue
- (2) Pneumonia
- (3) Tetanus
- (4) Plague

131. _____ is very effective sedative and painkiller and is very useful in patients who have undergone surgery ?

- (1) Marijuana
- (2) Nicotine
- (3) Cocaine
- (4) Morphine

132. Cancer cells are more easily damaged by radiation than normal cells because they are-

- (1) Starved of mutation
- (2) different in structure
- (3) Undergoing rapid division
- (4) Growth rate is slow

133. Most commonly used marker in clinical diagnosis of liver cancer is:-

- (1) Alpha fetoprotein
- (2) Alkaline phosphatase
- (3) Prostate specific antigen
- (4) Human chorionic gonadotrophin

134. BCG vaccine is a preventive measure against :-

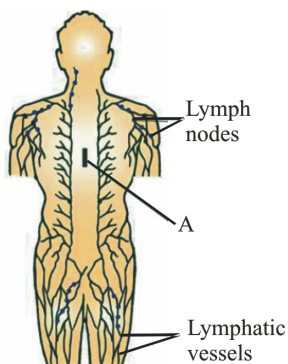
- (1) Typhoid
- (2) Tuberculosis
- (3) Whooping cough
- (4) Tetanus

135. Which of the following statement is incorrect ?

- (1) Heroin is chemically diacetylmorphine
- (2) LSD is obtained from fruiting body of fungus
- (3) With repeated use of drugs tolerance receptor level in our body decreases.
- (4) Atropa belladonna & datura shows hallucinogenic properties.

SECTION-B (BIOLOGY-I)

136. Which of the following statement is false for the structure labelled as " A " in the figure given below?



- (1) Maturation of lymphocytes takes place
(2) Immature lymphocytes differentiate into antigen- sensitive lymphocytes
(3) Lymphocytes interact with antigen and become effector cells
(4) Micro-environment is provided for development and maturation of T-lymphocytes.
137. How many of the following vaccines are included in live vaccine.
i-BCG, ii- MMR, iii-IPV, iv-OPV, v-Tetanus, vi-Rota virus, vii-Diphtheria, viii-Plague
(1) Five (2) One (3) Four (4) Three
138. Match the column-I with column-II and choose the correct options :-

Column-I (Disease)		Column-II (Antibodies formed against)	
(A)	Hashimoto's Disease	(i)	Castle intrinsic factor
(B)	Myasthenia gravis	(ii)	Thyroid gland
(C)	Pernicious Anemia	(iii)	β -cells of pancreas
(D)	Insulin dependent diabetes mellitus (I.D.D.M.)	(iv)	Acetylcholine receptors

- (1) A-iii, B-ii, C-iv, D-i (2) A-iv, B-iii, C-i, D-ii
(3) A-ii, B-iv, C-i, D-iii (4) A-ii, B-i, C-iv, D-iii

TG: @Chalnaayaaar

139. Tumor cells have been shown to avoid detection and destruction by immune system. Therefore the patient are given certain substances called biological response modifiers (BRM) which activates their immune system and helps in destroying the tumor. Identify the BRM :-

- (1) Alpha-interferon (2) Alpha lactalbumin
(3) Alpha-1 antitrypsin (4) Gamma globulin

140. **Statement-I :** Benign tumors *occasionally* remain confined to their original location and do not spread to other parts of the body.

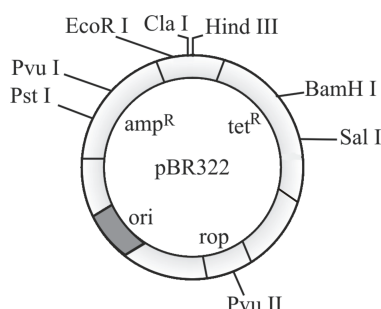
Statement-II: Metastasis is the most feared property of malignant tumor

- (1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

141. Which of the following tools of recombinant DNA technology is incorrectly paired with its use—

- (1) restriction enzyme - Production of RFLPs
(2) DNA ligase-that cuts DNA, creating the sticky ends
(3) DNA polymerase - used in a polymerase chain reaction to amplify section of DNA
(4) reverse transcriptase - production of cDNA from mRNA

- 142.



If the alien DNA is ligated at the Pvu-I site of ampicillin resistance gene in the given vector then gene which get inactivated due to insertion of alien DNA helps in the selection of :

- (1) Transformants (2) Recombinants
(3) Non-transformants (4) Both (2) and (3)

143. Restriction endonucleases :-

- (1) Are synthesized by bacteria as part of their defense mechanism
- (2) Are present in mammalian cells for degradation of DNA when the cell dies
- (3) Are used in genetic engineering for ligating two DNA molecules
- (4) Are used for invitro DNA synthesis

144. What % of recombinants produced in F_2 generation from cross I and cross II respectively conducted by Morgan on drosophila :-

- (1) 98.7 % and 1.3 %
- (2) 1.3 % and 37.2 %
- (3) 62.8 % and 37.2 %
- (4) 98.7 % and 62.8 %

145. How many type of gametes are produced by a pentahybrid plant (Tt Aa Bb Ee Gg) if T and A are completely linked gene :-

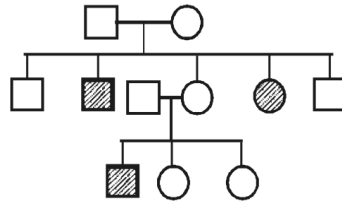
- (1) 32
- (2) 16
- (3) 8
- (4) 2

146. When a female parent transfers its X-linked character to the grand daughter through the offspring of opposite sex, it is known as :-

- (1) Criss-Cross inheritance
- (2) Non criss-cross inheritance
- (3) Holandric inheritance
- (4) More than one options are correct

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147.



In this given pedigree what is the mode of inheritance

- (1) Autosomal dominant
- (2) Autosomal recessive
- (3) X-linked dominant
- (4) X-linked recessive

148. Which one of the following sexually transmitted disease is incorrectly matched with its pathogen ?

- (1) Gonorrhea - Neisseria meningitides
- (2) Syphilis - Treponema pallidum
- (3) Trichomoniasis - Trichomona Vaginalis
- (4) Chlamydiasis - Chlamydia trachomatis

149. Match column I with column II and choose correct answer-

A.	Carcinoma	i.	initiation of new cancer
B.	Sarcoma	ii.	cancer of epithelial tissue
C.	Lymphoma	iii.	Bone, Cartilage tissue cancer
D.	Neoplasm	iv.	Hematopoietic cells tumor

- (1) A - iii, B - ii, C - i, D - iv
- (2) A - ii, B - iii, C - iv, D - i
- (3) A - ii, B - iv, C - iii, D - i
- (4) A - iii, B - i, C - ii, D - iv

150. Which of the following pair of disease is viral as well as transmitted by mosquitoes ?

- (1) Elephantiasis and dengue
- (2) Dengue and Ascariasis
- (3) Malaria and Chikungunya
- (4) Dengue and Chikungunya

Topic : GENETICS (POI) : CHROMOSOMAL THEORY OF INHERITANCE TO HUMAN GENETICS, BIOTECHNOLOGY (PART-I), DIGESTION AND ABSORPTION, EXCRETORY PRODUCTS AND THEIR ELIMINATION, [BACK UNIT] : HUMAN HEALTH AND DISEASE

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SECTION-A (BIOLOGY-II)

151. Hemophilia is more common in males because it is a :-

- (1) Recessive character carried by Y-chromosome
- (2) Dominant character carried by Y-chromosome
- (3) Recessive trait carried by X-chromosome
- (4) Dominant trait carried by X-chromosome

152. **Statement-I** : Colourblind person is unable to differentiate basics colours like red and green.

Statement-II : Colourblindness is an X-linked recessive disease which causes defect in cone cells.

- (1) Both statement I and II are correct.
- (2) Both statement I and II are wrong.
- (3) Statement I is correct and II is incorrect
- (4) Statement I is incorrect and II is correct.

153. In a population of 200 individuals, 140 belongs to genotype AA, 40 belongs to genotype Aa & 20 to aa. Find out frequency of "a" allele.

- (1) 0.2 (2) 0.4 (3) 0.6 (4) 0.8

154. **Statement-I** : Sex determination in honey bee is based on number of sets of chromosome an individual receives.

Statement-II : Female honey bee are haploid while males are diploid.

- (1) Both Statement-I & II are correct
- (2) Both Statement-I & II are incorrect
- (3) Statement-I is correct but statement II is incorrect
- (4) Statement-II is correct but statement I is incorrect

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PHASE - MEB

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155. Which of the following statement is incorrect ?

- (1) linked gene are non-allelic
- (2) Linked gene can be separated by crossing over.
- (3) linked gene are present on different chromosome
- (4) linkage term was given by T.H. Morgan.

156. Second and third letter in name of restriction endonuclease indicates :-

- (1) Genes of bacteria
- (2) Species of bacteria
- (3) Phylum of bacteria
- (4) Strain of bacteria,

157. Which of the following statement is incorrect ?

- (1) First restriction endonuclease discovered was Hind-II
- (2) Restriction endonuclease is an adaptation in bacteria against bacteriophage
- (3) Restriction endonuclease are known as molecular glue
- (4) Restriction site of recombinant DNA is made of 4 to 8 nucleotide with palindromic sequence

158. First recombinant DNA was prepared by :-

- (1) Karl Ereky (2) Paul Berg
- (3) Stanley Cohen (4) Kary Nullis

159. In order of transfer gene into animal cell which of the following vector is used ?

- (1) Ti-plasmid (2) PBR-322
- (3) λ -Bacteriophage (4) Retro virus

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160. Assertion (A) : In rDNA technology transformant with recombinant grow on medium containing ampicillin but not on tetracycline.

Reason (R) : Recombinant plasmid contain the gene of interest in tetracycline resistant gene.

- (1) (A) is not correct but (R) is correct.
- (2) Both (A) & (R) are correct and (R) is correct explanation of (A)
- (3) Both (A) & (R) are correct and (R) is not correct explanation of (A)
- (4) (A) is correct but (R) is correct.

161. Absence of enterokinase will affect which of the following process ?

- (1) Maltose $\rightarrow$ Glucose + Glucose
- (2) Procarboxypeptidase $\rightarrow$ Carboxypeptidase
- (3) Dipeptide $\rightarrow$ Amino acid
- (4) Starch $\rightarrow$ Maltose + α -Dextrin

162. Match column – I correctly with column – II

	Column – I		Column – II
(A)	deglutition	(i)	swallowing
(B)	digestion	(ii)	mechanical & biochemical method
(C)	bolus	(iii)	masticated food particles + saliva
(D)	chyme	(iv)	food mixed with gastric juice

- (1) A – (i), B – (ii), C – (iii), D – (iv)
- (2) A – (ii), B – (i), C – (iii), D – (iv)
- (3) A – (ii), B – (i), C – (iv), D – (iii)
- (4) A – (i), B – (ii), C – (iv), D – (iii)

163. Carbohydrates in the chyme are hydrolysed by ?

- (1) Salivary amylase (2) Pancreatic amylase
- (3) Gastric amylase (4) Bile juice

164. How many of the following enzyme is secreted by intestinal mucosa ?

Rennin, Trypsinogen, Enterokinase, Nuclease

- (1) 1
- (2) 2
- (3) 3
- (4) 4

165. Outer most layer of alimentary canal which is made up of mesothelium is called.

- (1) Muscularis
- (2) Sub-mucosa
- (3) Mucosa
- (4) Serosa

166. Which of the following is incorrect ?

- (1) Starch $\xrightarrow{\text{Amylase}}$ Maltose
- (2) Fats $\xrightarrow{\text{Lipases}}$ diglycerides
- (3) Nucleic acids $\xrightarrow{\text{Nucleotidase}}$ nucleotides
- (4) Lactose $\xrightarrow{\text{Lactase}}$ glucose + galactose

167. Digestion of proteins begins in the ___(i)___ and digestion of polysaccharides begins in the ___(ii)___ where (i) and (ii) are respectively :-

- (1) Mouth and stomach
- (2) stomach and small intestine
- (3) Stomach and mouth
- (4) Stomach and stomach

168. pH of salivary juice is :

- (1) 6.8
- (2) 1.8
- (3) 7.8
- (4) 3.8

169. Correctly match the components in column-I, II and III :-

Column-I		Column-II		Column-III	
(A)	Simple diffusion	a.	Glucose	i.	Concentration gradient, carrier protein
(B)	Facilitated diffusion	b.	Amino acid	ii.	Passive process
(C)	Active transport	c.	Fructose	iii.	Concentration gradient
(D)	Osmosis	d.	Water	iv.	Secondary active transport with Na^+ ions

(1) A-b-iii, B-c-i, C-a-iv, D-d-ii

(2) A-b-iii, B-c-iv, C-a-i, D-d-ii

(3) A-b-iii, B-c-i, C-a-ii, D-d-iv

(4) A-b-iii, B-b-i, C-a-iv, D-d-ii

170. Which is not true for brunner's gland ?

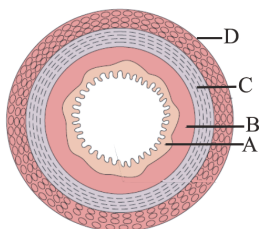
(1) It is multicellular gland

(2) It secretes non enzymatic part of intestinal juice

(3) It is found in submucosa of duodenum

(4) It opens in crypt of Leiberkuhn of sumucosa

171. Given below the diagram shows the transverse section of gut in which different parts are labelled by alphabets A,B,C,D. Choose the correct option in which all the alphabates are correctly labelled ?



	A	B	C	D
1	Serosa	Muscularis	Submucosa	Mucosa
2	Mucosa	Muscularis	Serosa	Submucosa
3	Mucosa	Submucosa	Muscularis	Serosa
4	Submucosa	Mucosa	Serosa	Musculari

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172. Which of the following part of nephron is not correctly matched with it's function ?

(1) **P.C.T** → All essential nutrient & 70 - 80% electrolytes reabsorbed

(2) **Ascending Limb of Loop of Henle** → Minimum Reabsorption

(3) **D.C.T.** → Conditional Reabsorption of Na^+ and H_2O

(4) **Collecting duct** → No role in maintainance of pH and Ionic balance of blood

173. **Statement-I** : Kwashiorkor is produced by protein deficiency unaccompanied by calorie deficiency.

Statement-II : One year and above child/children is/are, replaced with mother's milk by a high calorie low protein diet.

(1) Statement-I correct; statement-II incorrect

(2) Statement-I incorrect; statement-II correct

(3) Both statement-I and statement-II correct

(4) Both statement-I and statement-II incorrect

174. The activities of the gastro intestinal are under the control of :-

(1) Neural system

(2) Muscular system

(3) Hormonal system

(4) Both (1) and (3)

175. If loop of Henle is absent in mammalian nephron then urine :-

(1) Concentrated

(2) Diluted

(3) No change

(4) None of these

176. Which is reabsorbed actively in nephron?

- (a) Glucose (b) Na^+
(c) Amino acid (d) Nitrogenous wastes

- (1) a and b only (2) a, b and c
(3) a and d only (4) a, b and d

177. In mammals, which blood vessel would normally carry largest amount of urea ?

- (1) Renal Vein (2) Dorsal Aorta
(3) Hepatic Vein (4) Hepatic Portal Vein

178. Fill in the blanks in the following statement by choosing correct option from below :-

"Each kidney has nearly--- A---million complex---B---Structures called---C---which are the---D--
- units".

	A	B	C	C
(1)	1.5	tubular	nephron	structural
(2)	1-2	saccular	nephron	structural
(3)	1	tubular	nephron	functional
(4)	1	saccular	neuron	functional

179. The outline of principal event of urination is given below in unorder manner.

- (I) Stretch receptors on the wall of urinary bladder send signal to the CNS.
(II) The bladder fills with urine and becomes distended.
(III) Urination
(IV) CNS passes on motor messages to initiate the contraction smooth muscles of bladder and simultaneous relaxation of urethral sphincter.

The correct order of step for urination is :-

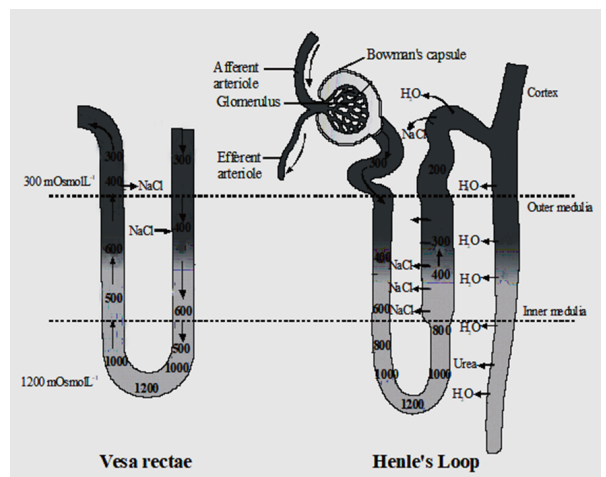
- (1) I → II → III → IV
(2) IV → III → II → I
(3) II → I → IV → III
(4) III → II → I → IV

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180. Select correct option with reference to two types of nephrons in kidneys :-

	Juxtamedullary Nephron	Cortical Nephron
(1)	Higher number	Lesser number
(2)	Loop of Henle longer	Loop of Henle shorter
(3)	Vasa recta less developed	Vasa recta more developed
(4)	No role in counter current mechanism	Maintenance of counter current mechanism

181. Which of the following statement is not correct about the following diagram



- (1) Mammals have the ability to produce concentrated urine
(2) The flow of filtrate in two limbs of Henle's loop is in same direction
(3) NaCl is transported by the ascending limbs of Henle's loop which is exchanged with descending limbs of vasa recta.
(4) Human kidney can produce urine nearly four times concentrated than the initial filtrate

SECTION-B (BIOLOGY-II)

182. JG-cells of juxta glomerular apparatus release:-

- (1) Rennin when fall in GFR
- (2) Angiotensinogen when blood flow decrease
- (3) Renin when fall in GFR
- (4) Aldosterone when fall in GFR

183. **Assertion** :- ADH (anti Diuretic Hormone) prevent diuresis.

Reason :- ADH facilitates water reabsorption from the latter parts of the tubule.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

184. Which of the following is totally reabsorbed in renal tubule?

- (1) Na^+ ions
- (2) K^+ ions
- (3) H_2O
- (4) $\text{C}_6\text{H}_{12}\text{O}_6$

185. In nephron, water reabsorption is maximum in :-

- (1) PCT
- (2) Loop of Henle
- (3) Glomerulus
- (4) DCT

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186. A test cross of F_1 progeny $\frac{+a}{+b}$ produces following offspring.

$$\begin{array}{l} \frac{++}{ab} = 16 \\ \frac{+b}{ab} = 35 \end{array} \quad \begin{array}{l} \frac{ab}{ab} = 14 \\ \frac{a+}{ab} = 35 \end{array}$$

What will be distance between linked gene :-

- (1) 16 CM
- (2) 14 CM
- (3) 30 CM
- (4) 35 CM

187. Which of the following relationship is incorrect.

- (1) Distance between linked gene $\downarrow$ - Recombination $\uparrow$
- (2) Age $\uparrow$ - linkage $\uparrow$
- (3) Crossing over $\uparrow$ - linkage $\downarrow$
- (4) Distance between linked gene $\uparrow$ - Crossing over $\uparrow$

188. The recombination frequency between the genes A & C is 25%, B & C is 9%, B & D is 31%, A & B is 16%, C & D is 40% and A & D is 15%. What will be sequence of these genes on a linear chromosome ?

- (1) CBAD
- (2) ACBD
- (3) DACB
- (4) CABD

189. Which method is used to check the progression of restriction enzyme ?

- (1) PCR
- (2) Gel Electrophoresis
- (3) Downstream processing
- (4) DNA finger printing

190. In polymerase chain reaction if DNA primers are not used then which of the following steps will be affected.

- (1) Denaturation and Annealing
- (2) Denaturation and Extension
- (3) Annealing and Extension
- (4) Only Denaturation

191. Emulsification of fats by bile takes place in-

- (1) Duodenum
- (2) Liver
- (3) Stomach
- (4) Large Intestine

192. Select the incorrect match pair

- (1) Jaundice-Liver is affected
- (2) Vomiting-Ejection of stomach contents
- (3) Constipation-Reduces the absorption of food.
- (4) Indigestion-Caused by inadequate enzymes secretion

193. Number of Incisor, Canine, Premolar and Molar in each jaw of human is :

- (1) 4, 2, 4, 6 (2) 4, 4, 6, 4
- (3) 6, 2, 4, 4 (4) 4, 2, 5, 6

194. Read the following statements (A-C) and answer the question following them.

A-Chylomicrons are very small proteins coated by fat globules.

B-Defaecation is a voluntary process mediated by mass peristalsis.

C-Inflammation of intestinal tract is the most common gut ailment.

How many of the above statements are correct?

- (1) None (2) One
- (3) Two (4) Three

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195. The physiological value of protein is :

- (1) 5.65 kcal/g
- (2) 9.45 kcal/g
- (3) 4.0 kcal/g
- (4) 9.0 kcal/g

196. Glucose and urea are present in :-

- (1) blood
- (2) Blood and urine
- (3) urine
- (4) Blood and glomerular filtrate

197. Match the column-I and column-II, choose the correct options.

	Column-I		Column-II
a	Amonotelic	i	Birds
b	Ureotelic	ii	Mammals
c	Uricotelic	iii	Aquatic animal

- (1) a-iii, b-ii, c-i
- (2) a-i, b-iii, c-ii
- (3) a-ii, b-i, c-iii
- (4) a-i, b-ii, c-iii

198. Which of the following are actively secreted into filtrate by tubular cells of nephron :-

- (1) Na^+ , K^+ , NH_3
- (2) K^+ , NH_3 , H^+
- (3) H^+ , HCO_3^+ , K^+
- (4) H^+ , NH_3 , Na^+

199. Aldosterone causes -

- (1) Excretion of Na^+ and reabsorption of H_2O
- (2) Reabsorption of Na^+ and excretion of H_2O
- (3) Reabsorption of Na^+
- (4) Excretion of H_2O and Cl^-

200. The reabsorption is minimum in which limb of Henle loop?

- (1) Descending limb
- (2) Ascending limb
- (3) Transvers limb
- (4) Sigmoid colon

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 ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MEB			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
07 13-08-2023	SYLLABUS COVERED FROM 17-07-2023 TO 12-08-2023	P	WAVE OPTICS, UNIT, DIMENSION & ERROR
		C	SOLUTION
		B	ORIGIN, EVOLUTION, DOMESTICATION OF ANIMALS

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